

Problem set #4

MATH 4503/6503: Numerical Methods for Differential Equations

Due date: March 25, 2026

Reading resources: Notes uploaded to D2L; book pages 13-32 and chapters 9, 10, and 11

Problem #1: PDEs (finally!)

Hello, people! I know students start getting real busy around this time of the semester, so I will keep this problem set short (just 3 problems instead of the usual 4). Just hang in there; we will have only one more problem set after this.

Let us start nice and slow on PDEs, letting Maple's inbuilt routines do the solving for us. Using `pdsolve/numeric`, create movies of $u(t, x)$ satisfying the so-called sine-Gordon equation (famous applications of which are seen in quantum field theory and differential geometry),

$$u_{tt} - u_{xx} + \sin(u), u(0, x) = f(x), u_t(0, x) = -f'(x), u(t, -50) = 0, u(t, 50) = 0,$$

for several choices of f . Here, subscripts denote partial differentiation and $t \in [0, 100]$.

Problem #2: diffusion+advection

The categories we classified PDEs into in class were way too idealized! Real world scenarios are much more complicated, and this problem serves as a little reminder of that. We will consider a situation involving diffusion and advection at the same time; therefore, the PDE for it cannot be put into either diffusion or advection category alone.

You are required to write a Maple procedure that uses a forward-time, centered-space stencil to solve the advection-diffusion equation

$$\frac{\partial u}{\partial t} = d \frac{\partial^2 u}{\partial x^2} + c \frac{\partial u}{\partial x}, d \in (0, \infty), c \in (-\infty, \infty)$$

subject to the boundary conditions

$$u(t, -1) = 1, u(t, 1) = 0.$$

The arguments of your procedure should include (but may not be limited to) initial data $f(x)$ for $u(t, x)$, the diffusion constant d , and the velocity parameter c . The output should be a movie.

Experiment with a few choices of initial data and simulation parameters. What could you say about the stability of this algorithm?

Problem #3: one-upping problem 2

Okay. You have guessed it! I could not come up with a better title for this problem. However, this is really what it is. We are trying to one-up the complexity of the real world expressed in problem 2 by considering a fourth order linear PDE for $u(t, x)$:

$$\frac{\partial u}{\partial t} = -\frac{\partial^4 u}{\partial x^4}.$$

We subject it to boundary conditions $u(t, \pm 1) = 0, u_x(t, \pm 1) = 0$. The subscript x here indicates partial differentiation. We introduce a spatial grid for a numerical solution to this problem as follows:

$$x_j = -1 + \frac{2j}{M+1}, h = x_{j+1} - x_j = \frac{2}{M+1},$$

where j is an integer and $M+1$ is the total number of steps from $x = -1$ to $x = 1$ (each with a step size of h). We will analyze the given equation using the method of lines we covered in the class, meaning that we start with the approximation

$$U_j(t) \approx u(t, x_j).$$

The first of the boundary conditions implies that

$$U_0(t) = U_{M+1}(t) = 0 \quad (*).$$

- Use a centered two-point stencil for the derivative in the second boundary condition to express $U_{-1}(t)$ in terms of $U_1(t)$ and $U_{M+2}(t)$ in terms of $U_M(t)$.
- If we use a centered five-point stencil for $\partial^4 u / \partial x^4$ in conjunction with $(*)$ and the results of part (a), we could approximate the given PDE as a matrix ODE:

$$\frac{d\mathbf{U}}{dt} = \mathbf{A}\mathbf{U}, \mathbf{U} = \begin{bmatrix} U_1(t) \\ U_2(t) \\ \vdots \\ U_M(t) \end{bmatrix}.$$

Write a procedure that takes M as an argument and returns the $M \times M$ matrix $\mathbf{M}$.

- The Crank-Nicolson stencil to solve the given PDE is equivalent to the trapezoidal stencil for the matrix ODE above with step size s . Write a Maple procedure to generate the Crank-Nicolson solution with arguments including initial data $f(x)$ for $u(t, x)$, the number of time steps N , the integer M , and the final time in the simulation τ . You can use `LinearSolve` to solve the linear system at each time step. The output of the procedure should be a movie.
- Compare your results to those generated by `pdsolve/numeric` assuming "top hat" initial

data

$$f(x) = \frac{\Theta(x + 1/2) + \Theta(x - 1/2)}{2},$$

where $\Theta(x - a)$ represents the Heaviside step function, which is implemented in Maple as `Heaviside(x-a)`.